

SULIT



SECP2523 DATABASE

SESSION 2024/2025 - SEMESTER 1

ALTERNATIVE ASSESSMENT REPORT: PHASE 2

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Case Study	Loan Management Module
System Name	KADA ESERVE
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SECTION C (SQL AND INTERFACE IMPLEMENTATION)

3.1 SQL Statement (DDL & DML)

3.1.1 Data Definition Language (DDL)

1.

```
CREATE TABLE `loans` (  
  `loan_id` varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL,  
  `member_id` bigint UNSIGNED NOT NULL,  
  `loan_type_id` bigint UNSIGNED NOT NULL,  
  `bank_id` bigint UNSIGNED NOT NULL,  
  `date_apply` date NOT NULL,  
  `loan_amount` double NOT NULL,  
  `interest_rate` double NOT NULL,  
  `monthly_repayment` double NOT NULL,  
  `monthly_gross_salary` double NOT NULL,  
  `monthly_net_salary` double NOT NULL,  
  `loan_period` int NOT NULL,  
  `status` varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'pending',  
  `created_at` timestamp NULL DEFAULT NULL,  
  `updated_at` timestamp NULL DEFAULT NULL  
)
```

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	loan_id 🔑	varchar(50)	utf8mb4_unicode_ci		No	None			Change Drop More
☐	2	member_id 🔑	bigint		UNSIGNED	No	None			Change Drop More
☐	3	loan_type_id 🔑	bigint		UNSIGNED	No	None			Change Drop More
☐	4	bank_id 🔑	bigint		UNSIGNED	No	None			Change Drop More
☐	5	date_apply	date			No	None			Change Drop More
☐	6	loan_amount	double			No	None			Change Drop More
☐	7	interest_rate	double			No	None			Change Drop More
☐	8	monthly_repayment	double			No	None			Change Drop More
☐	9	monthly_gross_salary	double			No	None			Change Drop More
☐	10	monthly_net_salary	double			No	None			Change Drop More
☐	11	loan_period	int			No	None			Change Drop More
☐	12	status	varchar(255)	utf8mb4_unicode_ci		No	pending			Change Drop More
☐	13	created_at	timestamp			Yes	NULL			Change Drop More
☐	14	updated_at	timestamp			Yes	NULL			Change Drop More

2.

```
CREATE TABLE `loan_types` (
  `loan_type_id` bigint UNSIGNED NOT NULL,
  `loan_type` varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL,
  `created_at` timestamp NULL DEFAULT NULL,
  `updated_at` timestamp NULL DEFAULT NULL
)
```

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action
☐	1	loan_type_id	bigint	UNSIGNED	No	None			Change Drop More
☐	2	loan_type	varchar(100)	utf8mb4_unicode_ci	No	None			Change Drop More
☐	3	created_at	timestamp		Yes	NULL			Change Drop More
☐	4	updated_at	timestamp		Yes	NULL			Change Drop More

3.1.2 Data Manipulation Language (DML)

<< Provide the SQL statements (DDL and DML) for a set of queries (minimum 10 queries) that can be implemented in your proposed module. In addition, please include the screenshots of the output/results for each SQL statement.>>

1.

```
INSERT INTO `member_register` (`id`, `created_at`, `updated_at`, `no_anggota`, `name`, `email`, `ic`, `phone`, `address`, `city`, `postcode`, `state`, `gender`, `DOB`, `agama`, `bangsa`, `no_pf`, `salary`, `office_address`, `office_city`, `office_postcode`, `office_state`, `guest_id`, `status`)
VALUES (1, '2025-01-24 16:38:36', '2025-01-24 16:39:08', '1234', 'gui', 'guisin@graduate.utm.my', '020304020202', '0129715408', 's', 'tangkak', '84900', 'Johor', 'Perempuan', '2025-01-03', 'Buddha', 'Cina', '344', '8000', 'vv', 'kelantan', '12345', 'Kelantan', 2, 'approved');
```

id	created_at	updated_at	no_anggota	name	email	ic	phone	address	city	postcode	state	gender	DOB	agama	bangsa	no_pf	salary	office_address	office_city	office_postcode	office_state	guest_id	status
1	2025-01-25 00:38:36	2025-01-25 00:39:08	1234	gui	guisin@graduate.utm.my	020304020202	0129715408	s	tangkak	84900	Johor	Perempuan	2025-01-03	Buddha	Cina	344	8000	vv	kelantan	12345	Kelantan	2	approved

2.

```
INSERT INTO `loans` (`loan_id`, `member_id`, `loan_type_id`, `bank_id`, `date_apply`, `loan_amount`, `interest_rate`, `monthly_repayment`, `monthly_gross_salary`, `monthly_net_salary`, `loan_period`, `status`, `created_at`, `updated_at`)
VALUES ('LOAN-1738035268', 1, 3, 1, '2025-01-28', 2888, 5, 247.23, 4000, 3600, 12, 'pending', NULL, NULL);
```

loan_id	member_id	loan_type_id	bank_id	date_apply	loan_amount	interest_rate	monthly_repayment	monthly_gross_salary	monthly_net_salary	loan_period	status	created_at	updated_at
LOAN-1738035267	1	3	1	2025-01-28	2888	5	247.23	4000	3600	12	approved	2025-01-28 11:34:27	2025-01-28 11:37:57
LOAN-1738035268	1	3	1	2025-01-28	2888	5	247.23	4000	3600	12	pending	NULL	NULL

3.

```
UPDATE Loans
SET status = 'approved'
WHERE loan_id = 'LOAN-1738035268';
```

loan_id	member_id	loan_type_id	bank_id	date_apply	loan_amount	interest_rate	monthly_repayment	monthly_gross_salary	monthly_net_salary	loan_period	status	created_at	updated_at
LOAN-1738035267	1	3	1	2025-01-28	2888	5	247.23	4000	3600	12	approved	2025-01-28 11:34:27	2025-01-28 11:37:57
LOAN-1738035268	1	3	1	2025-01-28	2888	5	247.23	4000	3600	12	approved	NULL	NULL

4.

```
DELETE FROM `loans`
WHERE loan_id = 'LOAN-1738035268';
```

loan_id	member_id	loan_type_id	bank_id	date_apply	loan_amount	interest_rate	monthly_repayment	monthly_gross_salary	monthly_net_salary	loan_period	status	created_at	updated_at
LOAN-1738035267	1	3	1	2025-01-28	2888	5	247.23	4000	3600	12	approved	2025-01-28 11:34:27	2025-01-28 11:37:57

5.

```
SELECT loans.loan_id, member_register.name, loans.loan_amount, loans.status
FROM loans
JOIN member_register ON loans.member_id = member_register.id;
```

```
mysql> SELECT loans.loan_id, member_register.name, loans.loan_amount, loans.status
-> FROM loans
-> JOIN member_register ON loans.member_id = member_register.id;
+-----+-----+-----+-----+
| loan_id | name | loan_amount | status |
+-----+-----+-----+-----+
| LOAN-1738035267 | gui | 2888 | approved |
| LOAN-1738035268 | gui | 2888 | pending |
+-----+-----+-----+-----+
2 rows in set (0.03 sec)
```

6.

```
SELECT loans.loan_id, loan_types.loan_type, banks.bank_name, loans.loan_amount
FROM loans
JOIN loan_types ON loans.loan_type_id = loan_types.loan_type_id
JOIN banks ON loans.bank_id = banks.bank_id
WHERE loans.status = 'approved';
```

```
mysql> SELECT loans.loan_id, loan_types.loan_type, banks.bank_name, loans.loan_amount
-> FROM loans
-> JOIN loan_types ON loans.loan_type_id = loan_types.loan_type_id
-> JOIN banks ON loans.bank_id = banks.bank_id
-> WHERE loans.status = 'approved';
+-----+-----+-----+-----+
| loan_id | loan_type | bank_name | loan_amount |
+-----+-----+-----+-----+
| LOAN-1738035267 | Skim Khas | Unknown Bank | 2888 |
+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

7.

```
SELECT loan_id, loan_amount, monthly_gross_salary
FROM loans
WHERE status = 'pending';
```

```
mysql> SELECT loan_id, loan_amount, monthly_gross_salary
-> FROM loans
-> WHERE status = 'pending';
+-----+-----+-----+
| loan_id | loan_amount | monthly_gross_salary |
+-----+-----+-----+
| LOAN-1738035268 | 2888 | 4000 |
+-----+-----+-----+
1 row in set (0.00 sec)
```

8.

```
SELECT loan_types.loan_type, SUM(loans.loan_amount) AS total_loan
FROM loans
JOIN loan_types ON loans.loan_type_id = loan_types.loan_type_id
GROUP BY loan_types.loan_type;
```

```
mysql> SELECT loan_types.loan_type, SUM(loans.loan_amount) AS total_loan
-> FROM loans
-> JOIN loan_types ON loans.loan_type_id = loan_types.loan_type_id
-> GROUP BY loan_types.loan_type;
+-----+-----+
| loan_type | total_loan |
+-----+-----+
| Skim Khas | 5776 |
+-----+-----+
1 row in set (0.01 sec)
```

3.2 Interface Implementation

1. Loan application

Dashboard Ahli Permohonan Pinjaman Status Pinjaman Laporan Individu Laporan Tahunan gui

Permohonan Pinjaman

1 Maklumat Peribadi 2 Maklumat Pinjaman 3 Maklumat Penjamin 4 Pengesahan

Maklumat Peribadi

Nama
gui

Alamat
s

Email

No. Kad Pengenalan
020304020202

No. Telefon
0129715408

2. Approve loan application

KADA Admin

Dashboard

Ahli

Permohonan Baru

Kewangan

Laporan Tahunan

Senarai Pinjaman

Pengurusan pinjaman ahli

+ Kadar Faedah

Papar 10 entri

Cari:

ID PINJAMAN	NAMA PEMINJAM	JENIS PINJAMAN	JUMLAH (RM)	TEMPOH	STATUS	TINDAKAN
LOAN-1738035267	gui	Skim Khas	2,888.00	12 bulan	Approved	
LOAN-1738035268	gui	Skim Khas	2,888.00	12 bulan	Approved	

Memaparkan 1 hingga 2 daripada 2 entri

Sebelumnya1Seterusnya

3. Check loan status

Dashboard Ahli

Permohonan Pinjaman

Status Pinjaman

Laporan Individu

Laporan Tahunan

gui

PINJAMAN

Permohonan Dihantar

Telah Diproses

Diluluskan

AMAUN PINJAMAN	BAYARAN BULANAN	KADAR	TARIKH PINJAMAN	TINDAKAN
RM 2888	RM 247.23	5%	2025-01-28 11:34:27	BUTIRAN EXPORT

Permohonan Dihantar

Sedang Diproses

Belum Disahkan

AMAUN PINJAMAN	BAYARAN BULANAN	KADAR	TARIKH PINJAMAN	TINDAKAN
RM 2888	RM 247.23	5%		BUTIRAN EXPORT

Pinjaman Skim Khas

EXPORT

KEMBALI

SEDANG DIPROSES

Pinjaman ID	:LOAN-1738035268	Gaji Bersih	:RM 3,600.00
Amaun	:RM 2,888.00	Gaji Kasar	:RM 4,000.00
Bayaran Bulanan	:RM 247.23	Kadar	:5%
Tempoh	:12 bulan	Tarikh Pinjaman	:

Nama Bank

: Unknown Bank

Akaun No

: 12345678

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SECTION D (CONCLUSION AND REFLECTION)

4.1 Conclusion

The proposed Loan Management Module of KADA-eServe has successfully addressed the shortcomings of the previous manual system by introducing a centralized, fully digitalized, and database-driven system. This module optimizes loan application process, enhances KADA staff convenience, and significantly reduces the risk of human error and manpower. Members benefit from seamless application processes with pre-filled forms, real-time status tracking, and easy access to repayment details. For admin also known as clerk, features such as automated notifications, real-time updates, and secure data storage improve operational efficiency, allow for better scalability, and reduces workloads.

The integration of this module with other components of KADA-eServe, such as membership and reporting, ensures smooth workflows and streamlined operations across the board. By implementing database normalization, the system reduces data redundancy, ensures consistency, and improves overall system reliability. Furthermore, the relational database schema also highlights the focus on data integrity and long-term sustainability.

4.2 Reflection

First and foremost, our WBL project focused on developing a system from scratch into a fully functional website, integrating knowledge from software engineering, databases as well as system development technology courses. My role primarily focused on developing the annual report module, which included functionalities for uploading and saving thumbnail and PDF files in a public repository and storing their paths in the database. Each team member was respectively assigned specific modules, and we often collaborated to complete certain management features. We conducted brainstorming sessions, iterative development, and testing to meet the project goals. Despite some miscommunication and delays with my module due to conflicts, the team also maintained a good collaboration overall, with everyone fulfilling their responsibilities effectively.

At the time, I felt quite stressed and anxious as I encountered challenges while developing my own module, especially when the code conflicts caused delays. However, I managed to resolve these problems eventually, which brought a sense of relief and accomplishment. The stress sometimes impacted my focus, but the supportive collaboration among teammates and their suggestions helped ease the pressure and got the clue to solve it. The major difficulty for me was timing, as resolving issues often required additional time, but it was ultimately overcome through my perseverance and help from my teammates.

Reflecting on this experience, I selected this incident because it tested my problem-solving abilities and highlighted the importance of seeking help within a team. The challenges I faced in handling the conflicts were a valuable lesson, as they showed me how crucial it is to keep calm under pressure and reach out for assistance when necessary. Looking forward, I recognize the need to improve my time management and seek help to avoid delays in the future.

As for what remains, I still have questions about how I can better manage my time and anticipate potential problems, especially when dealing with error, tight deadlines and complex tasks. I plan to refine my approach and create early planning for time management and work on strategies to streamline collaboration in future projects.

For the future, I intend to be more proactive in planning by allocating time for each task based on my own abilities. I will focus on setting realistic goals that align with my strengths when working with tight deadlines, ensuring I can manage them effectively within the given timeframe. For tasks with more flexibility, I will take on more challenging sections to push myself further. By evaluating my own capacity and breaking tasks into manageable chunks, I aim to avoid the stress that comes with looming deadlines. Additionally, I will prioritize maintaining clear communication within the team to stay on track and ask for help when needed, but always with the understanding that I can rely on my planning skills to handle tasks within the available time.

